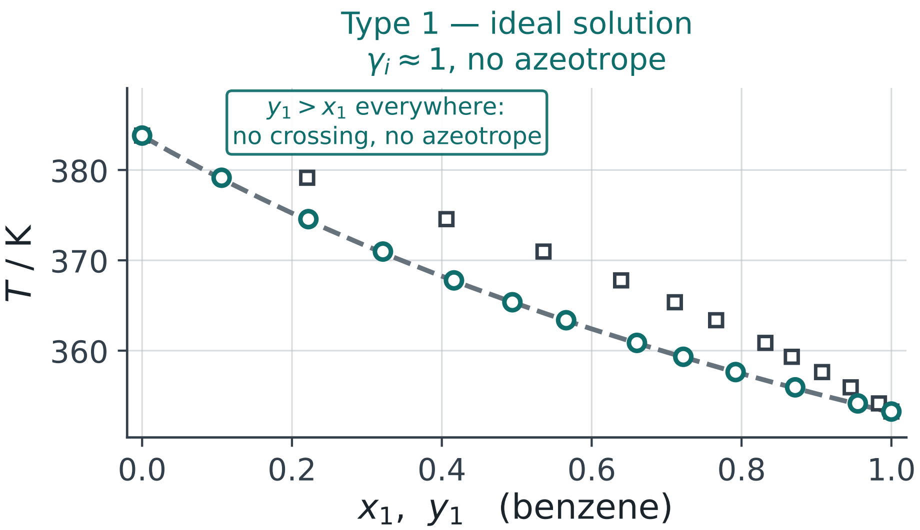
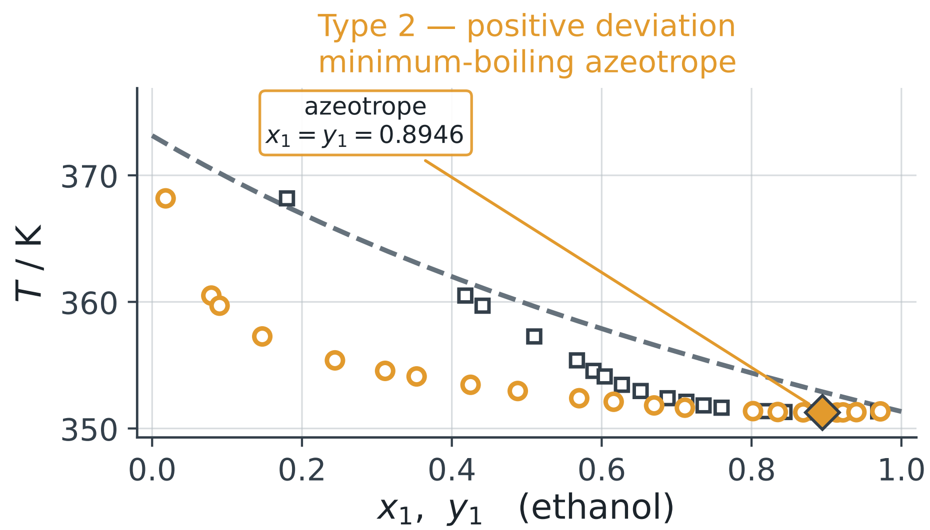
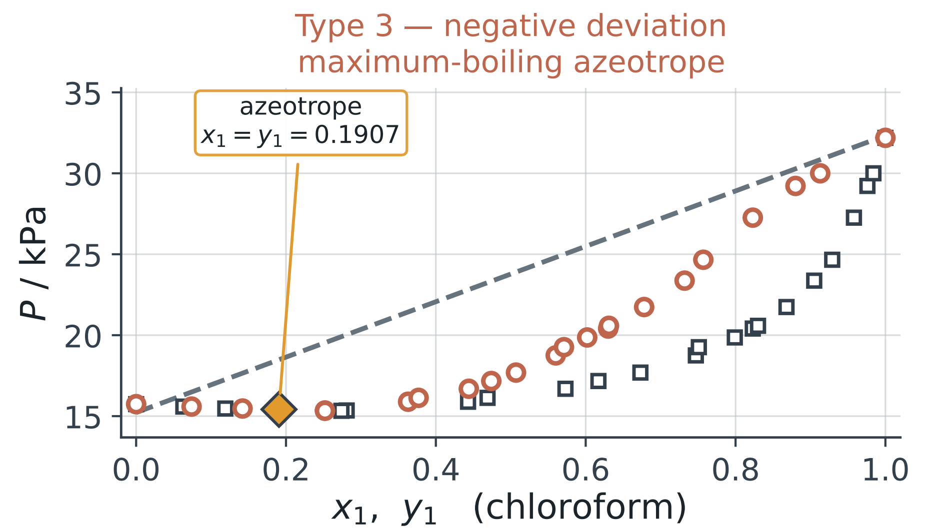
The four solution types, on five published datasets — the dashed line is Raoult's law with the same Antoine constants throughout



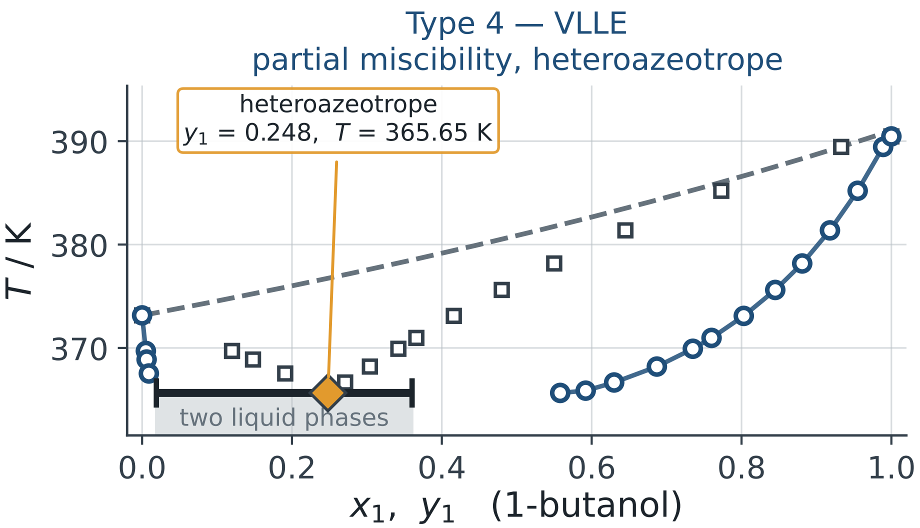
benzene (1) + toluene (2), isobaric, P = 101.33 kPa, 13 points
γ_1 = 0.990–1.003, γ_2 = 0.934–1.014
largest departure from Raoult: +0.17 K at x_1 = 0.222
doi:10.1016/j.fluid.2013.12.002



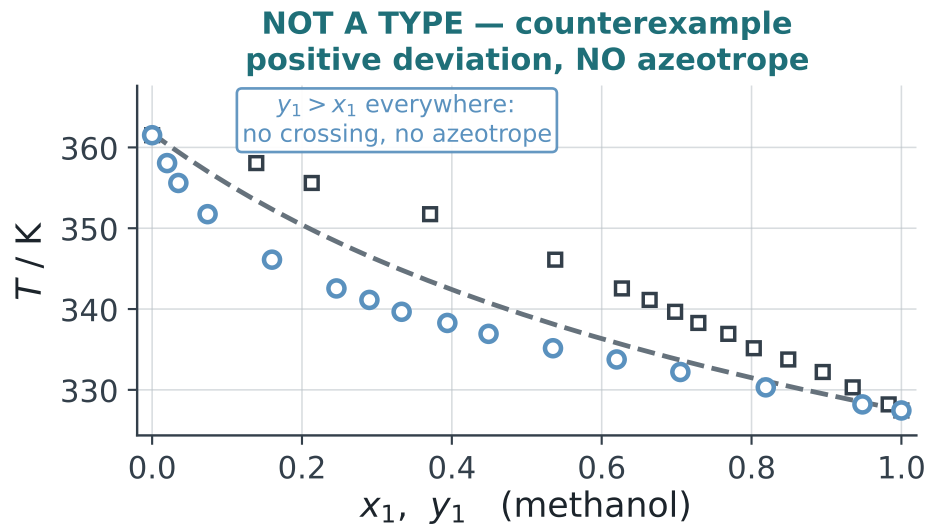
ethanol (1) + water (2), isobaric, P = 101.3 kPa, 21 points
γ_1 = 0.997–5.303, γ_2 = 0.999–2.543
largest departure from Raoult: −11.19 K at x_1 = 0.147
doi:10.1021/je2008704



chloroform (1) + 2-butanone (2), isothermal, T = 303.15 K, 22 points
γ_1 = 0.401–1.002, γ_2 = 0.362–1.043
largest departure from Raoult: −26.86 % at x_1 = 0.444
doi:10.1021/je060150a



1-butanol (1) + water (2), isobaric, P = 101.3 kPa, 17 points
γ_1 = 0.997–57.117, γ_2 = 1.000–3.501
largest departure from Raoult: −16.64 K at x_1 = 0.592
conjugate liquids x_1 = 0.0190 and 0.3603; gap 0.009–0.558
doi:10.1021/je700411p



methanol (1) + water (2), isobaric, P = 66.52 kPa, 16 points
γ_1 = 1.007–2.166, γ_2 = 1.008–1.371
largest departure from Raoult: −6.24 K at x_1 = 0.160
doi:10.1021/je300613v

How to read these five panels

The dashed line on every panel is Raoult's law, evaluated with the same Antoine constants as the data. Everything between the points and that line is G^E, and nothing else.

- Type 1 sits on the line.
- Types 2 and 4 sit below it (isobaric) — $\gamma_i > 1$.
- Type 3 sits above the line in T, below it in P — $\gamma_i < 1$.
- An azeotrope is where $y_1 - x_1$ changes sign. It is a consequence of the size of G^E, not a fifth category.

The counterexample is the point of the fifth panel: methanol + water deviates positively, γ_1 reaches 2.17, and it still has no azeotrope. Positive deviation does NOT imply an azeotrope.

TYPE 4 IS A DIFFERENT QUESTION. Whether one liquid phase or two is a matter of stability, not of azeotropy. The flat line at 365.65 K on the Type 4 panel is a three-phase invariant: two liquids and one vapour, all fixed in composition. Reading an azeotrope off it by interpolating across the gap gives x_1 = 0.2121, which is fiction — there is no liquid at that composition.

VLLE IS TREATED PROPERLY IN MODULE 5, where the tangent-plane criterion decides how many phases there are before any of this is attempted. Module 4 can only show that the split is there.